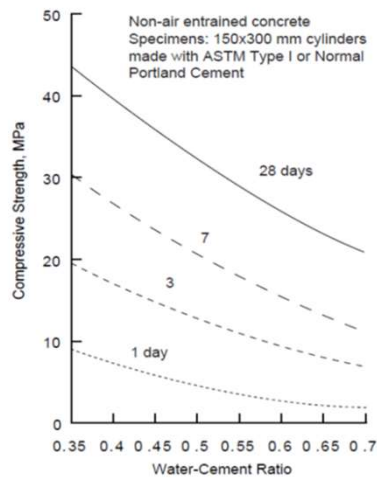
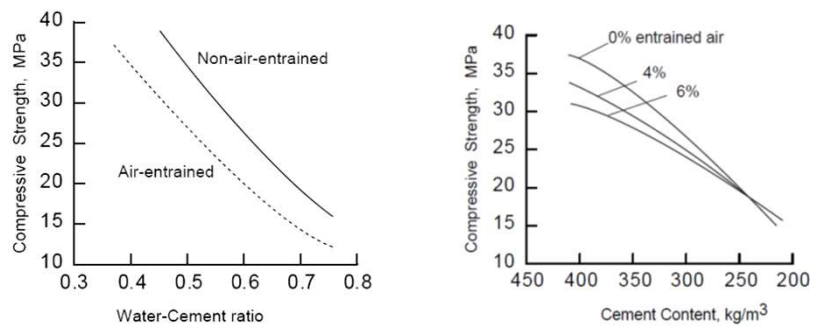


Compressive strength of concrete



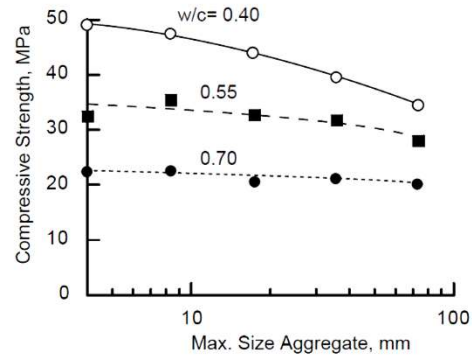
Influence of w/c ratio and moist curing age on compressive strength of concrete

Compressive strength of concrete



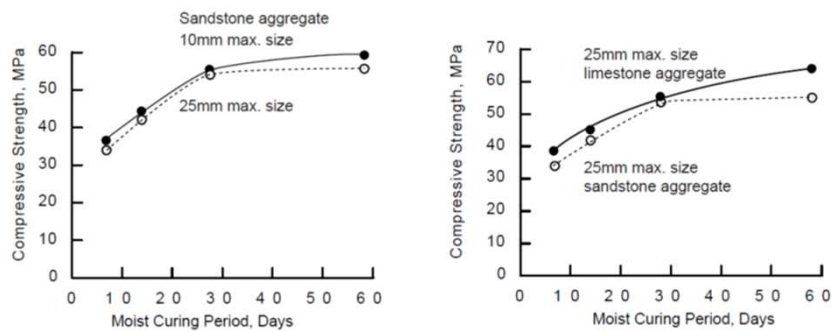
Influence of w/c ratio, entrained air and cement content on compressive strength of concrete

Compressive strength of concrete



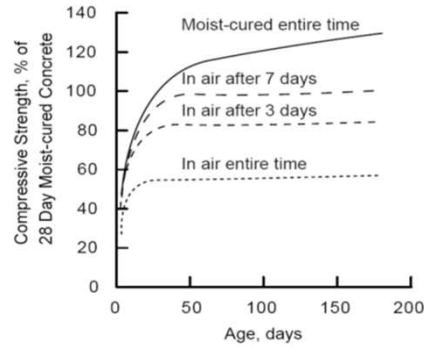
Influence of aggregate size and w/c ratio on compressive strength of concrete

Compressive strength of concrete



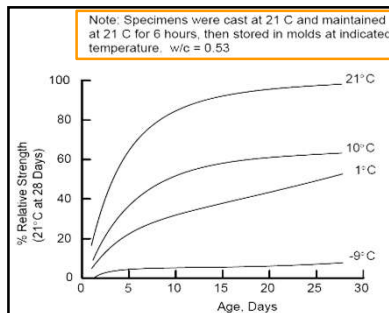
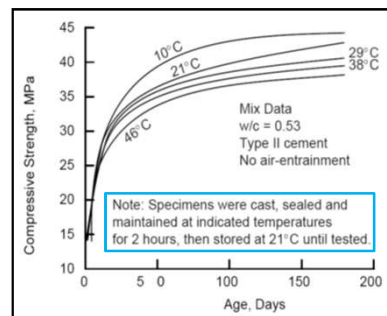
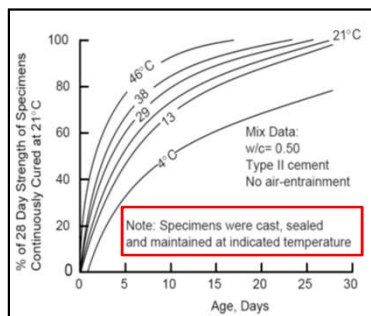
Influence of aggregate size and mineralogy on compressive strength of concrete

Compressive strength of concrete

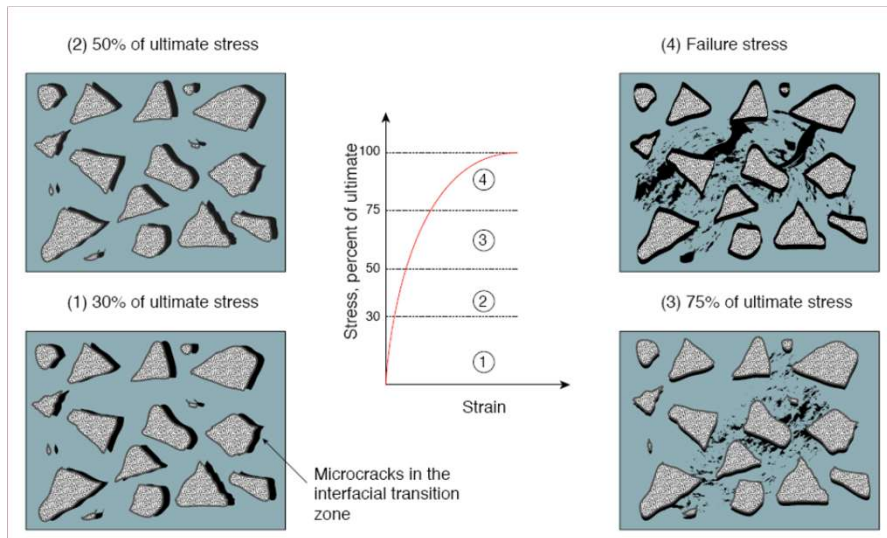


Influence of curing conditions on compressive strength of concrete

Compressive strength of concrete

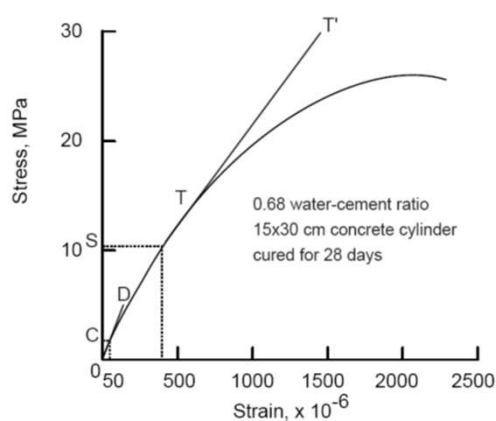


Influence of casting and curing temperatures on compressive strength of concrete



Diagrammatic representation of stress-strain behaviour of concrete under uniaxial compression

Different types of elastic moduli



Tangent modulus: slope of the line TT' drawn tangent to any point on the stress-strain curve

Secant modulus: slope of the line corresponding to stress **SO**

Chord modulus: slope of the line corresponding to stress **SC**

Dynamic modulus (initial tangent modulus): slope of the line **OD** from the origin

Tangent modulus:

- It is given by the slope of a line drawn tangent to the stress-strain curve at any point on the curve.

Secant modulus:

- It is given by the slope of a line drawn from the origin to a point on the stress-strain curve corresponding to 40% stress of the failure load.

Chord modulus:

- It is given by the slope of a line drawn from a point representing a longitudinal strain of 50 $\mu\text{m}/\text{m}$ to the point that corresponds to 40% of the ultimate load. Shifting the base line by 50 microstrain is recommended to correct for the slight concavity that is often observed at the beginning of stress-strain curve.

Dynamic modulus:

- It corresponds to a very small instantaneous strain and is approximately given by the initial tangent modulus i.e., tangent modulus for a line drawn at the origin.
- For stress analysis of structures subjected to earthquake or impact loading, it is more appropriate to use dynamic modulus of elasticity, which can be determined more accurately by a sonic test.

Curing of concrete

- ✓ Curing is the process of maintaining satisfactory moisture content and temperature in the concrete for a definite period of time.
- ✓ Hydration of cement is a long-term process, and requires water and proper temperature. Thus, curing allows continued hydration and consequently, continued gain in compressive strength of concrete.

- Curing can be performed by any of the following methods:

i) Maintaining the presence of water in the concrete during early age:

Methods include ponding, immersion, spraying or fogging, and wet coverings.

ii) Preventing the loss of mixing water from the concrete by sealing the surface:

Methods include impervious papers or plastic sheets, membrane-forming compounds, and leaving the forms in place.

iii) Accelerating the strength gain by supplying heat and additional moisture to the concrete:

Methods include steam curing, insulating blankets or covers, and various heating techniques such as electrical, hot oil and infrared curing.

Shrinkage of concrete

- ✓ Shrinkage is caused by the loss of water due to:
 - evaporation
 - hydration of cement
 - carbonation
- ✓ In practice, shrinkage is measured as a linear strain.

Plastic shrinkage

- While the cement paste is in plastic state, it undergoes a volumetric contraction. This contraction is known as plastic shrinkage. The magnitude of the volumetric contraction is of the order of 1% of the absolute volume of dry cement.
- Plastic shrinkage is caused due to the loss of water by evaporation from the surface of concrete or by suction by dry concrete below.
- The contraction induces tensile stress in the surface layers of concrete because they are restrained by the non-shrinking inner concrete and since, the concrete is very weak in plastic state, plastic cracking can readily occur at the surface.

Plastic shrinkage

- Plastic shrinkage is greater when the rate of evaporation of water is higher, which in turn depends on air temperature, concrete temperature, relative humidity of air, and wind speed.
- A complete prevention of evaporation immediately after casting reduces the plastic shrinkage of concrete.
- Since, it is the loss of water from the plastic cement paste that is responsible for plastic shrinkage, it is greater when the cement content of the mix higher, and lower when the aggregate content (by volume) is larger.

Autogenous shrinkage

- Autogenous shrinkage occurs even when there is no moisture movement to or from the set concrete.
- It is caused by the loss of water consumed in the hydration reaction of cement. Autogenous shrinkage is very small. Except in massive concrete structures, it is not distinguished from the shrinkage of hardened concrete due to loss of water to the outside.

Swelling

- If there is continuous supply of water to the concrete during hydration, concrete expands due to absorption of water by the cement gel. This process is known as swelling. Swelling is smaller than shrinkage of concrete.

Drying shrinkage

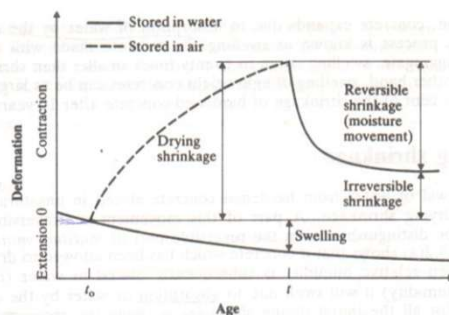
- It is caused due to withdrawal of water from the hardened concrete stored in unsaturated air. A part of this movement is irreversible.

- Not all the initial drying shrinkage is recovered even after prolonged storage in water.

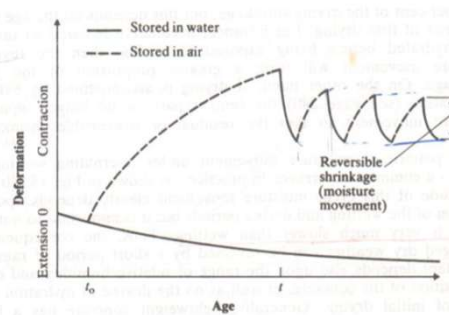
- For the usual range of concretes, the reversible moisture movement or wetting expansion represents about 40% to 70% of the drying shrinkage. However, this depends on the age before the onset of first drying.

- If the concrete is cured so that it is fully hydrated before being exposed to drying, then the reversible moisture movement will form a greater proportion of drying shrinkage. On the other hand, if drying is accompanied by extensive carbonation, the cement paste is no longer capable of moisture movement, which leads to higher residual or irreversible shrinkage.

- Irreversible shrinkage is associated with the formation of additional physical and chemical bonds in the cement gel when the adsorbed water has been removed.



(a)



(b)

Moisture movement in concrete:

(a) concrete which has dried from age " t_0 " until age " t " and was then re-saturated

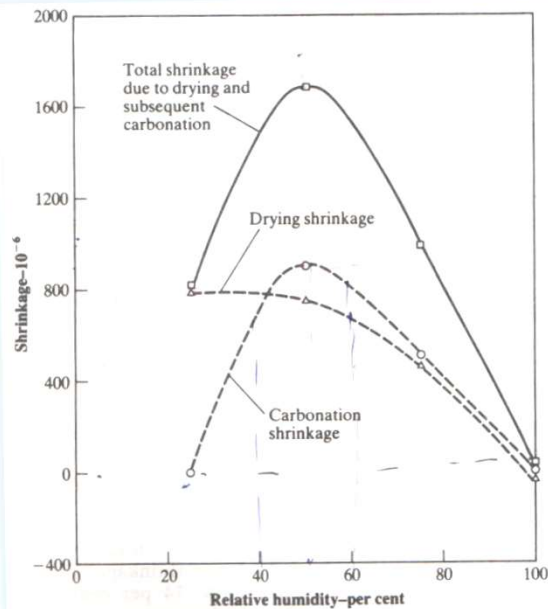
(b) concrete which has dried from age " t_0 " until age " t " and was then subjected to cycles of drying and wetting

Carbonation shrinkage

- Diffusion of CO_2 into concrete.
- CO_2 gets dissolved in concrete pore solution and results in formation of carbonic acid, which reacts with $\text{Ca}(\text{OH})_2$ to form CaCO_3 .
- A concomitant of the process of carbonation is a contraction of concrete known as carbonation shrinkage.
- Carbonation proceeds from the surface of concrete inwards, however at a very slow rate.

Actual rate of carbonation depends on:

- Permeability and moisture content of concrete
- CO_2 content, and relative humidity of ambient medium



Drying shrinkage and carbonation shrinkage of mortar at different values of relative humidity

References

- Mehta, P. K., and Monteiro, P. J. M. Concrete: Microstructure, Properties, and Materials, Tata McGraw-Hill, Third edition, 2006.
- Mamlouk, M. S., and Zaniewski, J. P. Materials for Civil and Construction Engineers, Pearson, Prentice Hall, Second edition, 2006.
- Neville, A. M., and Brooks, J. J. Concrete Technology, Pearson Education, Fourth Indian reprint, 2004.